

# Machine Learning 1

## Mock Exam A

Duration: 2 hours      Total: 100 points

Name: \_\_\_\_\_

Student ID: \_\_\_\_\_

### Instructions

- Show all important intermediate steps. Answers without justification can receive reduced credit.
- State any additional assumptions that you use.
- Give matrix and vector dimensions where the question asks for them.
- This course uses numerator layout: if  $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$ , then its Jacobian has shape  $n \times m$ .
- Unless a question states otherwise, vectors are column vectors.
- The suggested working times total 110 minutes. Reserve about 10 minutes to check your answers.

### Exercise 1: Multivariate derivatives and regularization 20 points, 18 minutes

#### 1.1 Temperature-scaled log-softmax

Let  $x \in \mathbb{R}^d$  and let  $\tau > 0$ . Define  $s : \mathbb{R}^d \rightarrow \mathbb{R}^d$  and  $\ell : \mathbb{R}^d \rightarrow \mathbb{R}^d$  by

$$s_i(x) = \frac{\exp(x_i/\tau)}{\sum_{k=1}^d \exp(x_k/\tau)}, \quad \ell_i(x) = \log s_i(x).$$

1. Derive  $\partial \ell_i / \partial x_j$  using index notation. Express the answer using  $s_j$ ,  $\tau$ , and the Kronecker delta  $\delta_{ij}$ . [5 points]
2. Write the full Jacobian  $J_\ell(x)$  in matrix notation and state its dimensions. [4 points]
3. Show that  $J_\ell(x)\mathbf{1} = 0$ , where  $\mathbf{1}$  is the all-ones vector. Explain this result using a transformation of  $x$  that leaves  $\ell(x)$  unchanged. [4 points]

#### 1.2 Generalized ridge regression

Let  $X \in \mathbb{R}^{n \times p}$ ,  $y \in \mathbb{R}^n$ ,  $\theta \in \mathbb{R}^p$ , and  $\gamma > 0$ . Let  $R \in \mathbb{R}^{p \times p}$  be symmetric positive definite. Consider

$$L(\theta) = \frac{1}{n} \|X\theta - y\|_2^2 + \gamma \theta^\top R \theta.$$

4. Derive the stationary point of  $L(\theta)$ . Then show that it is the unique global minimizer, even when  $X$  does not have full column rank. [7 points]

**Exercise 2: Waiting times and Bayesian estimation** 24 points, 25 minutes

A monitoring system checks a component repeatedly. Each check succeeds independently with probability  $q$ , where  $0 < q < 1$ . Let  $K$  be the number of failed checks before the first success. Its probability mass function is

$$p(K = k | q) = q(1 - q)^k, \quad k = 0, 1, 2, \dots$$

You may use

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1 - r}, \quad |r| < 1.$$

1. Derive  $\mathbb{E}[K | q]$ . Do not only state the result. [3 points]
2. Compute  $P(K = 2 | q = 0.25)$ . Give an exact or decimal value. [3 points]

Now suppose that  $K_1, \dots, K_N$  are independent observations and define  $S = \sum_{n=1}^N K_n$ . Assume  $S > 0$ .

3. Write the likelihood and log-likelihood of  $q$ , up to constants that do not depend on  $q$ . [3 points]
4. Derive the maximum-likelihood estimator of  $q$ . Verify that your stationary point is a maximum. [4 points]

Place a Beta( $a, b$ ) prior on  $q$ , with density

$$p(q) \propto q^{a-1}(1 - q)^{b-1}, \quad 0 < q < 1,$$

where  $a > 1$  and  $b > 1$ .

5. Derive the posterior distribution of  $q$  and identify its parameters. [4 points]
6. Derive the MAP estimator of  $q$ . [4 points]
7. Identify a one-dimensional statistic of the data that contains all data dependence in the posterior. Briefly justify your answer. [3 points]

**Exercise 3: Estimating an unknown range** 14 points, 14 minutes

Measurements  $X_1, \dots, X_N$  are independent and uniformly distributed on  $[-M, M]$ , where  $M > 0$  is unknown. Define

$$A = \max_{1 \leq n \leq N} |X_n|.$$

Assume that at least one observed value is nonzero, so  $A > 0$ .

1. Write the likelihood  $p(x_1, \dots, x_N | M)$ , including the support constraint. [4 points]
2. Derive the maximum-likelihood estimator of  $M$ . For the observations

$$-3.2, \quad 1.1, \quad 2.4, \quad -2.8, \quad 0.5,$$

give its numerical value.

[3 points]

You may use

$$\mathbb{E}[A] = \frac{NM}{N+1}, \quad \mathbb{E}[A^2] = \frac{NM^2}{N+2}.$$

3. Derive a multiplicative correction of the maximum-likelihood estimator that is unbiased for  $M$ .  
[3 points]
4. Derive the variance of your unbiased estimator and simplify the result. [4 points]

#### Exercise 4: Multi-output regression with correlated noise 23 points, 29 minutes

For each observation, let  $\phi_n \in \mathbb{R}^m$  be a feature vector and  $t_n \in \mathbb{R}^k$  be a target vector. Let  $W \in \mathbb{R}^{m \times k}$ . The model is

$$t_n = W^\top \phi_n + \varepsilon_n, \quad \varepsilon_n \sim \mathcal{N}(0, \sigma^2 C),$$

where  $C \in \mathbb{R}^{k \times k}$  is known and symmetric positive definite, while  $\sigma^2 > 0$  is unknown. The noise vectors are independent.

Let  $\Phi \in \mathbb{R}^{N \times m}$  have rows  $\phi_n^\top$ , and let  $T \in \mathbb{R}^{N \times k}$  have rows  $t_n^\top$ . Assume that  $\Phi$  has full column rank,  $N > m$ , and the data cannot be fitted exactly.

1. State the dimensions of  $\Phi W$ ,  $T - \Phi W$ , and  $(T - \Phi W)^\top (T - \Phi W)$ . [2 points]
2. Derive the log-likelihood of  $W$  and  $\sigma^2$ , including all terms that depend on either parameter. You may express the quadratic term using a trace. [6 points]
3. Derive the maximum-likelihood estimator of  $W$ . Show why it does not depend on  $C$  or  $\sigma^2$ . [7 points]
4. Using the fitted value of  $W$ , derive the maximum-likelihood estimator of  $\sigma^2$ . [5 points]
5. Let  $E = T - \Phi W_{\text{ML}}$ . Show that  $\Phi^\top E = 0$  and explain the geometric meaning of this identity. [3 points]

#### Exercise 5: Information from one Bayesian observation 19 points, 24 minutes

After observing a data set  $\mathcal{D}_N$ , the posterior over the regression weights is

$$p(w \mid \mathcal{D}_N) = \mathcal{N}(w \mid \mu_N, \Sigma_N),$$

where  $w, \mu_N \in \mathbb{R}^m$  and  $\Sigma_N \in \mathbb{R}^{m \times m}$  is symmetric positive definite. A new observation has feature vector  $\phi \in \mathbb{R}^m$ , target  $t \in \mathbb{R}$ , and likelihood

$$p(t \mid w, \phi) = \mathcal{N}(t \mid w^\top \phi, \beta^{-1}),$$

where  $\beta > 0$  is the observation-noise precision.

The standard precision and natural-parameter updates are

$$\begin{aligned} \Sigma_{N+1}^{-1} &= \Sigma_N^{-1} + \beta \phi \phi^\top, \\ \Sigma_{N+1}^{-1} \mu_{N+1} &= \Sigma_N^{-1} \mu_N + \beta \phi t. \end{aligned}$$

You may use the Sherman–Morrison identity

$$(A + uv^\top)^{-1} = A^{-1} - \frac{A^{-1}uv^\top A^{-1}}{1 + v^\top A^{-1}u}.$$

1. Derive the rank-one covariance update

$$\Sigma_{N+1} = \Sigma_N - \frac{\Sigma_N \phi \phi^\top \Sigma_N}{\beta^{-1} + \phi^\top \Sigma_N \phi}.$$

[4 points]

2. Derive the innovation form of the mean update

$$\mu_{N+1} = \mu_N + K(t - \phi^\top \mu_N),$$

and give the gain vector  $K$ . Explain the roles of the direction and the scalar residual in this update. [5 points]

3. Before observing  $t$ , derive the posterior predictive distribution  $p(t \mid \phi, \mathcal{D}_N)$ . Give both its mean and variance, and identify the two sources of predictive variance. [4 points]

4. For any direction  $a \in \mathbb{R}^m$ , derive

$$a^\top (\Sigma_N - \Sigma_{N+1}) a.$$

Show that it is nonnegative. State exactly when it is zero and explain what this says about the parameter directions learned from the observation. [6 points]

**End of Mock Exam A**